NOVALEARN

A project report submitted in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF COMPUTER APPLICATION

BY

KEERTHY SAJI [230021077981]

Under the guidance of

Mrs. ALEENA KUNJUMON

DEPARTMENT OF COMPUTER APPLICATIONS

BASELIOS POULOSE SECOND COLLEGE

PIRAMADOM

Affiliated to

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

APRIL 2026

BASELIOS POULOSE SECOND COLLEGE

PIRAMADOM

(Affiliated to Mahatma Gandhi University)

DEPARTMENT OF COMPUTER APPLICATIONS

Certificate

*This is to certify that the project entitled "**NovaLearn**" was submitted in partial fulfilment of the requirements for the award of the degree of BACHELOR OF COMPUTER APPLICATION is a bonafide report of the project done by Keerthy Saji (Reg no: 230021077981) during the year 2025-26.*

Internal Guide	**Head of Department**	**Principal**
Mrs. Aleena Kunjumon	**Mr. Bibu Skaria**	**Mrs. Tintu Mathew**

Submitted for the University Exam held on …….

Internal Examiner	**External Examiner**

DECLARATION

*I **Keerthy Saji** [230021077981] hereby declare that the project report, titled "NOVALEARN" is a record of original work undertaken by me for the award of the degree of Bachelor of Computer Application. I have completed this project under the guidance of **Mrs.ALEENA KUNJUMON,** Department Of Computer Applications. I also declare that this project has not been submitted for the award of any degree. I hereby confirm the originality of the work.*

Place: Name:

Date: Sign:

ACKNOWLEDGEMENT

I express our sincere thanks to **Mrs. TINTU MATHEW**, principal of Baselios Poulose Second College, Piramadom, for provided us with an opportunity to carryout work.

I am very grateful to **Mr.BIBU SKARIA** Head of the department of computer Application, Baselios Poulose Second College Piramadom and to all other teaching and nonteaching staffs in our college.

I express my profound gratitude to my guide, **Mrs.ALEENA KUNJUMON**, Assistant Professor of the Department of computer Application, Baselios Poulose Second College, Piramadom ,for the unfailing and valuable guidance and whole hearted supervision for the conduct of my dissertation.

I express my sincere thanks to our guide, **Mrs.MINU N MOHAN ,**staff of Gentrius Solutions, for giving valuable information about our project.

I also remember with great gratitude for all the encouragement given by my parents, family members, well-wishers and my dear friends in the successful completion of my work. Above all I are thankful to **ALMIGHTY GOD** for successful completion of my work.

ABSTRACT

"**NovaLearn**" is a comprehensive web-based Learning Management System (LMS) designed to provide an efficient and interactive digital learning environment for students, teachers, and administrators. The platform aims to streamline course management, content delivery, performance evaluation, and payment handling within a unified system, ensuring a structured and user-friendly educational experience.

The system is built around three primary user roles — Admin, Teacher, and Student — each with specific functionalities to ensure smooth operation. The Admin is responsible for managing users, courses, sections, and batches. Admins can assign teachers to specific batches, monitor student enrollments, track payment details, and oversee system-wide activities through a centralized dashboard. The admin also controls exam processes and ensures that only eligible students can participate based on predefined criteria.

The Teacher module enables instructors to upload study materials such as videos, notes, and PDFs, assign and evaluate assignments, and monitor student performance. Teachers can review submitted assignments, provide marks, and analyze overall progress through performance dashboards. The system also allows teachers to access relevant academic and payment-related information for better course management.

The Student module provides access to course materials, assignments, and performance tracking through a personalized dashboard. Students can enroll in courses, complete payments, submit assignments, and monitor their academic progress. The platform includes a structured eligibility mechanism where students must complete required payments and assignment submissions before being permitted to attend examinations.

NovaLearn incorporates an automated evaluation system that combines assignment scores and examination results to generate a final performance score. Based on this, the system generates digital certificates with unique identification codes, ensuring authenticity and easy verification.

Overall, NovaLearn serves as a scalable, secure, and efficient LMS that enhances digital learning by integrating course management, performance tracking, eligibility control, and certification within a single platform, thereby improving the quality and accessibility of education.

CONTENTS

SL.NO	TITLE	PAGE NO
1	**INTRODUCTION**	**8**
2	**SYSTEM STUDY** 2.1 EXISTING SYSTEM 2.1.1 DRAWBACKS 2.1.2 SYSTEM ANALYSIS 2.1.2.1 IDENTIFYING NEEDS OF SYSTEM 2.2 PROPOSED SYSTEM 2.2.1 ADVANTAGES OF PROPOSED SYSTEM 2.3 FEASIBILITY STUDY 2.3.1 ECONOMIC FEASIBILITY 2.3.2 TECHNICAL FEASIBILITY 2.3.3 BEHAVIOURAL FEASIBILITY	**9** 9 9 10 10 11 11 12 12 12 13
3	**SYSTEM SPECIFICATION** 3.1 ABOUT FRONT END 3.2 ABOUT BACK END 3.3 ABOUT OPERATING SYSTEM	**14** 14 15 16
4	**SYSTEM ANALYSIS AND DESIGN** 4.1 DATA FLOW DIAGRAM 4.2 TABLE DESIGN	**17** 17 24
5	**SYSTEM DESCRIPTION**	**33**

6	**SYSTEM TESTING AND IMPLEMENTATION** 6.1 SYSTEM TESTING 6.1.1 UNIT TESTING 6.1.2 INTEGRATION TESTING 6.1.3 VALIDATION TESTING 6.1.4 OUTPUT TESTING 6.2 SYSTEM IMPLEMENTATION	**36** 36 36 36 37 37 37
7	**CONCLUSION**	**38**
8	**BIBLIOGRAPHY**	**39**
9	**APPENDIX** 9.1 INPUT FORMS 9.2 OUTPUT FORMS	**40** 40 43

INTRODUCTION

NovaLearn is an innovative and comprehensive digital learning platform designed to connect students, teachers, and administrators within a single interactive environment. With a user-centric approach, the system simplifies course management, content delivery, performance tracking, and academic evaluation, creating a modern and structured online learning experience.

The administrative features empower the Admin to oversee and manage all system activities effectively. Admins can manage student and teacher accounts, organize courses into sections and batches, assign teachers to specific batches, and monitor overall system performance. The platform also enables the admin to handle payment tracking, manage course enrollments, and control examination processes.

For Students, NovaLearn provides an interactive space to access learning resources and track academic progress. Students can register, enroll in courses, access study materials such as videos, notes, and PDFs, and submit assignments assigned by teachers. The platform includes a structured workflow where students must complete course payments and submit required assignments before becoming eligible to attend examinations. Students can also view their marks, monitor progress through dashboards, and download certificates with unique identification codes upon successful course completion.

Teachers benefit from dedicated features that support content creation and student evaluation. They can upload learning materials, assign tasks, and assess student submissions efficiently. Teachers can monitor individual and batch-wise performance, provide marks for assignments and examinations, and contribute to the overall evaluation process. The system also allows teachers to track student engagement and academic progress through visual dashboards.

One of the key features of NovaLearn is its integrated evaluation and certification system. The platform automatically combines assignment scores and examination results to generate a final performance score. Based on this evaluation, digital certificates with unique IDs are generated, ensuring authenticity and easy verification

In essence, NovaLearn represents a modern approach to digital education by integrating course management, performance evaluation, payment handling, and certification into a single platform. Through its well-structured modules and user-friendly design, the system enhances accessibility, transparency, and efficiency in the learning process, creating a reliable and scalable solution for online education.

SYSTEM STUDY

2.1 EXISTING SYSTEM

In the field of education, traditional and partially digital learning systems are widely used for delivering knowledge and managing academic activities. These systems include physical classrooms, standalone learning platforms, and basic online tools. However, most of these systems operate in isolation and lack integration between course delivery, performance tracking, and administrative management.

2.1.1 Drawbacks of Existing System:

1. **Traditional Classroom System:**
   Learning is restricted to physical presence, making it difficult for students to access materials anytime. There is limited flexibility, and tracking student performance manually is time-consuming.

2. **Separate Learning Platforms:**
   Many platforms provide only specific features such as content sharing or assignment submission, but lack integration with payments, certification, and performance tracking.

3. **Manual Evaluation System:**
   Assignment correction and result calculation are often done manually, leading to delays and chances of human error.

4. **Lack of Centralized Management:**
   Student records, course details, payments, and performance data are managed separately, causing inefficiency and poor coordination.

5. **Limited Accessibility:**
   Students cannot access study materials, assignments, and results in a single place, reducing convenience and learning efficiency.

6. **No Structured Eligibility System:**
   There is no proper mechanism to ensure that students complete required tasks (like assignments or payments) before attending examinations.

7. **Poor Communication:**
   Interaction between students, teachers, and administrators is not streamlined, leading to confusion and delays.

2.1.2 System Analysis

2.1.2.1 Identifying the Needs of the System:

1. **Unified Learning Platform:**
   A centralized system is required to integrate course management, content delivery, assignments, exams, and payments.
2. **Automated Evaluation:**
   The system should automatically calculate results by combining assignment and exam marks to reduce manual effort.
3. **Secure User Management:**
   Role-based access for Admin, Teacher, and Student must be implemented to ensure proper control and security.
4. **Structured Workflow:**
   A system is needed where students must complete payments and assignments before becoming eligible for exams.
5. **Performance Tracking:**
   Dashboards with graphical representations should be included to monitor student progress and academic performance.
6. **Digital Certification:**
   Provide certificates with unique identification codes for authenticity and easy verification.
7. **Accessible Interface:**
   The platform should be user-friendly and accessible across devices for better usability.
8. **Efficient Communication:**
   Enable smooth interaction between students, teachers, and administrators.
9. **Payment Integration:**
   Secure and transparent payment handling system for course enrollment.
10. **Scalability:**
    The system should support a growing number of users, courses, and data efficiently.

2.2 PROPOSED SYSTEM

In response to the limitations of existing systems, the proposed system — NovaLearn — introduces a centralized, secure, and user-friendly Learning Management System. It integrates course management, content delivery, assignment handling, performance evaluation, payment processing, and certification into a single platform, ensuring an efficient and structured learning environment.

2.2.1 Advantages of Proposed System:

1. **All-in-One Learning Platform:**
   Students, teachers, and admins can manage all academic activities within a single system.
2. **Role-Based Access Control:**
   Separate dashboards and functionalities for Admin, Teacher, and Student ensure better management and security.
3. **Automated Evaluation System:**
   Assignment and exam marks are automatically combined to generate final results.
4. **Structured Eligibility Mechanism:**
   Students must complete payments and assignments before attending exams, ensuring discipline.
5. **Interactive Dashboards:**
   Graphical representation of performance, progress, and system statistics improves decision-making.
6. **Secure Payment System:**
   Students can make payments safely and access receipts anytime.
7. **Digital Certification:**
   Certificates with unique IDs are generated for verification and authenticity.
8. **Efficient Content Management:**
   Teachers can upload and manage study materials easily.
9. **Centralized Data Management:**
   All data related to users, courses, and performance is stored in one system.
10. **Scalable and Accessible:**
    The platform can handle multiple users and is accessible anytime from anywhere.

2.3 FEASIBILITY STUDY

A feasibility study was conducted to evaluate the practicality and effectiveness of the proposed system. The study determines whether the system is economically, technically, and behaviorally feasible.

2.3.1 ECONOMIC FEASIBILITY

Economic feasibility evaluates whether the benefits of the system outweigh its costs. NovaLearn is developed using cost-effective technologies such as Python Django, MySQL, and Bootstrap, which are open-source and widely supported. The system reduces manual work, improves efficiency, and minimizes operational costs, making it economically beneficial.

2.3.2 TECHNICAL FEASIBILITY

Technical feasibility analyzes whether the required technology and resources are available to implement the system. NovaLearn uses reliable technologies such as Python Django for backend processing, MySQL for database management, and HTML, CSS, JavaScript, and Bootstrap for frontend development. These technologies are well-supported and suitable for building scalable web applications.

2.3.2.1 Hardware Requirements

- Processor: Pentium IV or above
- RAM: 256 MB or above
- Hard Disk: 500 MB or above

2.3.2.2 Software Requirements

- Operating System: Windows 7 or above
- Web Browser: Any modern browser
- Technologies Used:
  - Python Django (Backend)
  - MySQL (Database via XAMPP)
  - HTML, CSS, JavaScript, Bootstrap

2.3.3 BEHAVIOURAL FEASIBILITY

Behavioral feasibility focuses on user acceptance of the system. NovaLearn is designed with a simple and user-friendly interface, making it easy for students, teachers, and administrators to use without requiring advanced technical skills. The system improves efficiency, reduces manual workload, and enhances user experience, ensuring high acceptance among users.

SYSTEM SPECIFICATIONS

ABOUT THE FRONT END

The front end of the NovaLearn system is developed using HTML, CSS, JavaScript, jQuery, Bootstrap, and AJAX. These technologies are used to create an interactive, responsive, and user-friendly interface that allows smooth interaction between users and the system. They help in designing dashboards, forms, course pages, and other user interfaces in a structured and visually appealing manner.

HTML

HTML (HyperText Markup Language) is the standard language used to create and structure web pages. It consists of a set of elements and tags that define the layout of a webpage, including headings, paragraphs, forms, tables, and links. HTML forms are widely used in web applications to collect user input such as login details, course enrollments, and assignment submissions.

In NovaLearn, HTML is used to design the basic structure of all pages including student dashboards, teacher panels, and admin interfaces. It ensures proper organization of content and enables easy navigation across different modules of the system. Its simplicity and compatibility with all browsers make it a reliable technology for web development.

CSS

CSS (Cascading Style Sheets) is used to control the presentation and layout of web pages. It allows developers to apply styles such as colors, fonts, spacing, and alignment to HTML elements, ensuring a consistent look and feel across the entire application. CSS also supports responsive design, making web pages adaptable to different screen sizes.

In NovaLearn, CSS is used to enhance the visual appearance of dashboards, forms, and course pages. It improves readability and user experience by creating a clean and organized interface. By separating design from structure, CSS makes the system easier to maintain and update.

JavaScript

JavaScript is a client-side scripting language used to create dynamic and interactive web applications. It enables real-time interaction between the user and the system by handling events, validating forms, and updating content without reloading the page.

In NovaLearn, JavaScript is used for form validation, interactive dashboards, and improving user experience. It ensures that user inputs are validated before submission and helps in creating a more responsive and efficient system.

jQuery

jQuery is a lightweight JavaScript library designed to simplify scripting tasks such as event handling, DOM manipulation, and AJAX operations. It reduces the amount of code required and improves development efficiency while ensuring cross-browser compatibility.

In NovaLearn, jQuery is used to handle user interactions, validate forms, and implement dynamic features within the interface. It works along with JavaScript to make the application more interactive and responsive, especially in handling real-time updates.

Bootstrap

Bootstrap is a popular front-end framework used for designing responsive and mobile-first web applications. It provides pre-built components such as buttons, forms, navigation bars, and grid systems, which help in developing consistent and visually appealing interfaces quickly.

In NovaLearn, Bootstrap is used to design responsive dashboards and layouts that work seamlessly across devices like desktops, tablets, and smartphones. It improves the overall look of the system and reduces development time.

AJAX

AJAX (Asynchronous JavaScript and XML) is a technique used to communicate with the server without refreshing the entire web page. It allows data to be sent and received asynchronously, making the application faster and more interactive.

In NovaLearn, AJAX is used to perform operations such as loading data dynamically, submitting forms, and updating content in real-time. This improves system performance and provides a smoother user experience.

ABOUT THE BACK END

The back end of the NovaLearn system is developed using Python Django, with MySQL as the database and XAMPP as the local server environment. These technologies work together to handle data processing, business logic, and database management efficiently

Python Django

Django is a high-level web framework built using Python that enables rapid development of secure and scalable web applications. It follows the Model-View-Template (MVT) architecture, which separates application logic, user interface, and data management. Django also provides built-in features such as authentication, admin panel, and security mechanisms.

In NovaLearn, Django is used to handle core functionalities such as user management, course handling, assignments, payments, and result processing. It ensures data security and efficient system performance, making it suitable for building a robust LMS.

MySQL

MySQL is an open-source relational database management system used to store and manage structured data. It organizes data into tables and supports SQL queries for efficient data retrieval and manipulation.

In NovaLearn, MySQL is used to store data related to users, courses, assignments, marks, and payments. It ensures data integrity, consistency, and reliability while handling multiple users simultaneously.

XAMPP Server

XAMPP is a free and open-source server environment used for local development and testing of web applications. It provides tools to manage databases and run server-side applications easily.

In NovaLearn, XAMPP is used to manage the MySQL database and provide a local server setup during development. It simplifies testing and debugging before deploying the system.

ABOUT THE OPERATING SYSTEM

The OS used is Windows Operating System.

WINDOWS OS

The system is developed and tested on the Windows Operating System, which provides a stable and user-friendly environment for web development. It supports all necessary tools such as web browsers, development frameworks, and database servers required for building the application.

Windows OS ensures smooth execution of both front-end and back-end components of NovaLearn. Its compatibility and ease of use make it an ideal platform for developing, testing, and running web-based applications.

SYSTEM ANALYSIS AND DESIGN

System design's main aim is to identify the modules that should be in the system, and the specifications of these modules and how they interact with each other to produce the desired results. At the end of the system design all the major data structures, file formats and the major modules in the system and their specification are decided.

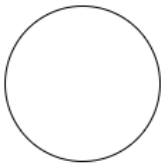
DATA FLOW DIAGRAM

A DFD has the purpose of clarifying system requirements and identifying major transformation that will become programs in system design.

These symbols are used in the DFD**.**

Source or destination of data

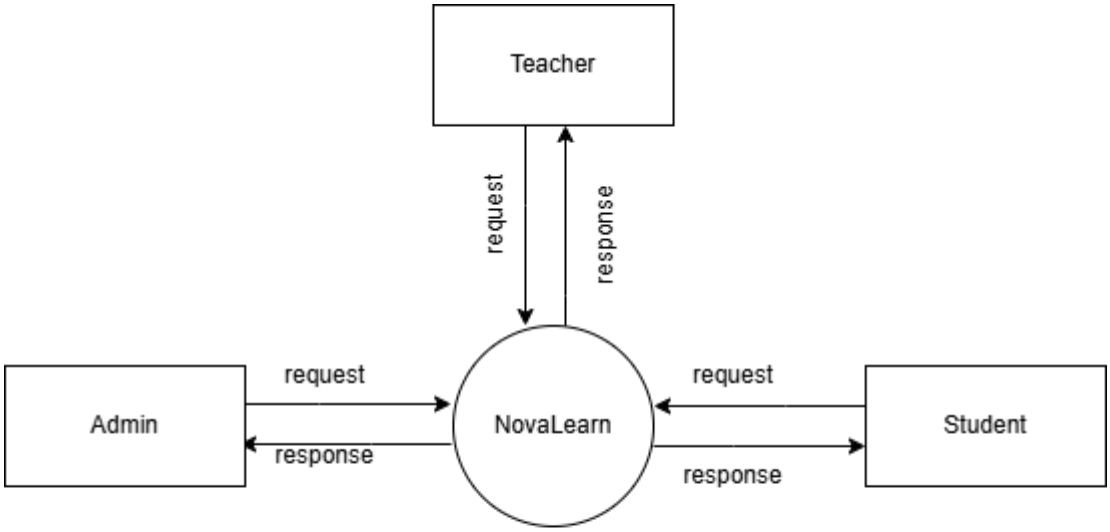
Data Flow



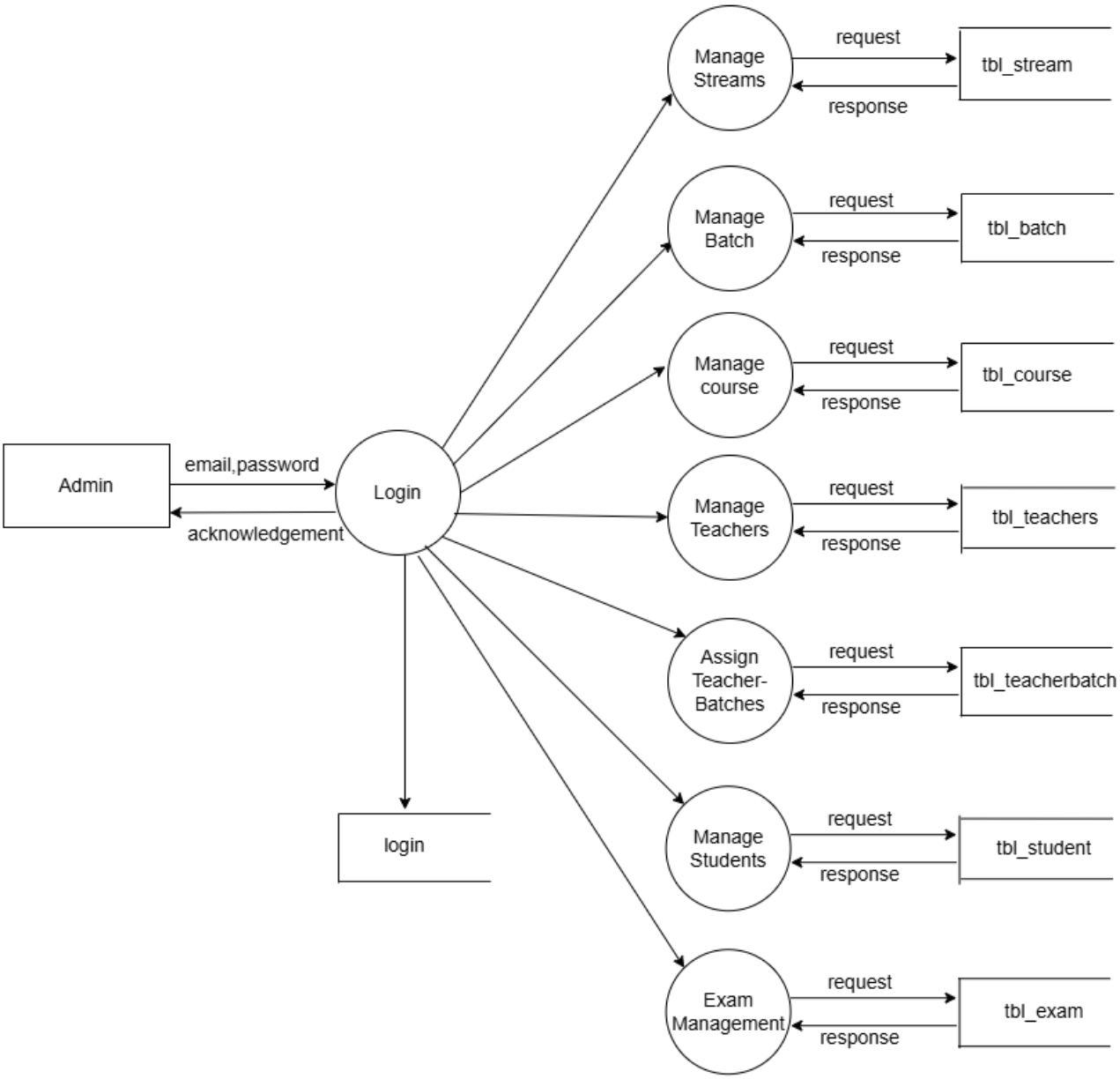
Process that transforms data flow

Data store

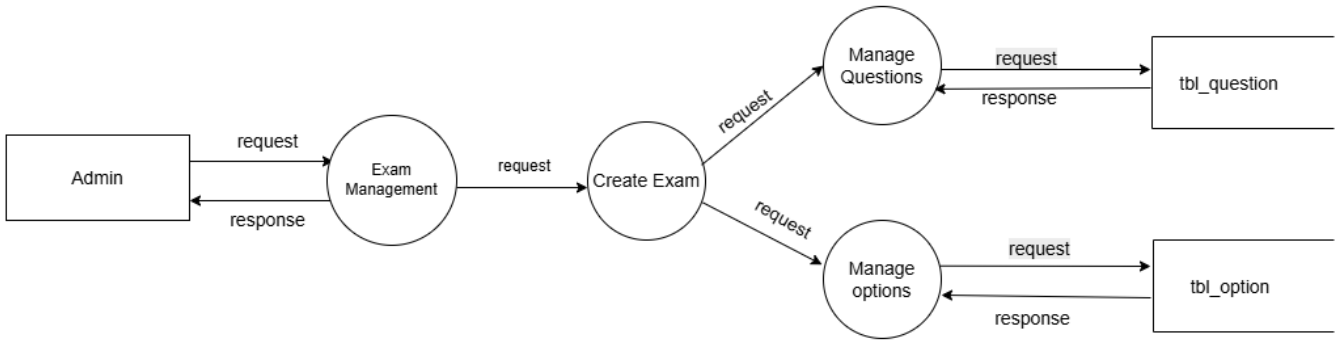
CONTEXT LEVEL DFD



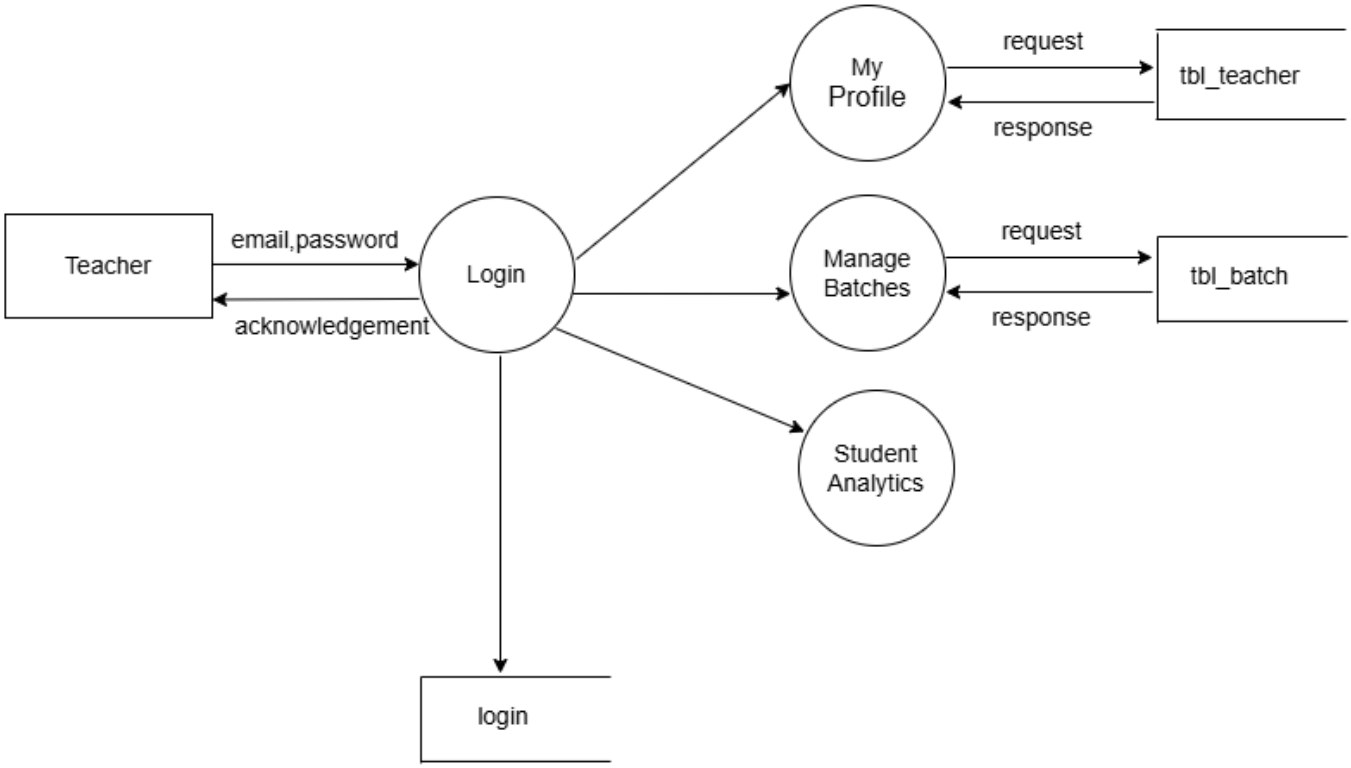
LEVEL 1.0 DFD


Admin
email,password
acknowledgement
Login
login
Manage Streams
request
response
tbl_stream
Manage Batch
request
response
tbl_batch
Manage course
request
response
tbl_course
Manage Teachers
request
response
tbl_teachers
Assign Teacher-Batches
request
response
tbl_teacherbatch
Manage Students
request
response
tbl_student
Exam Management
request
response
tbl_exam

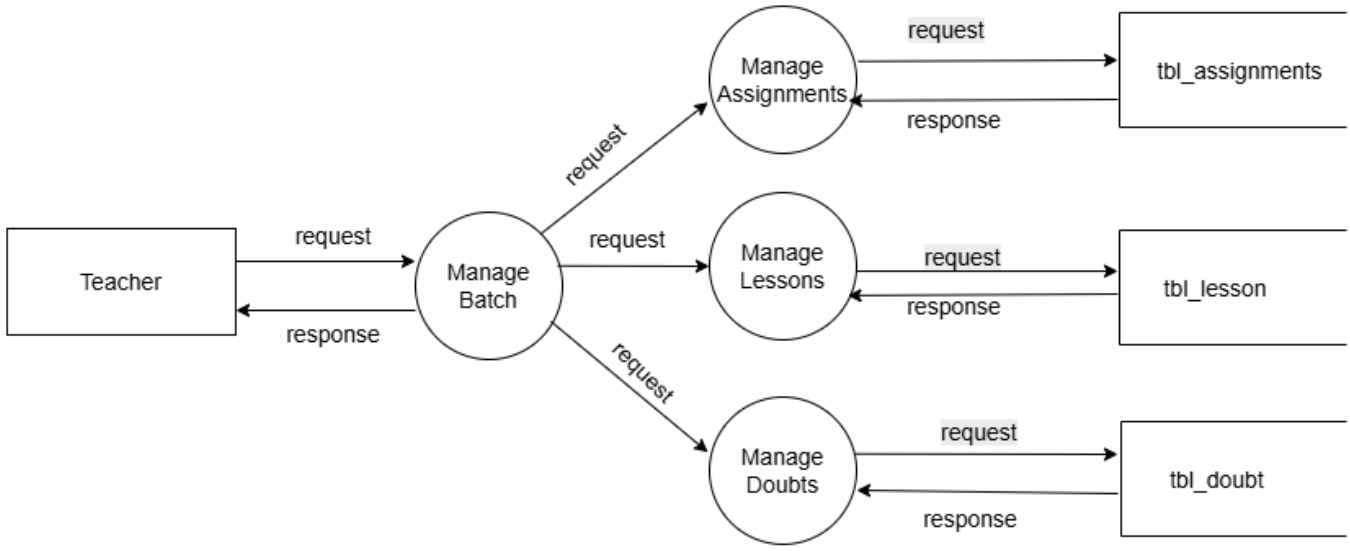
LEVEL 1.1 DFD



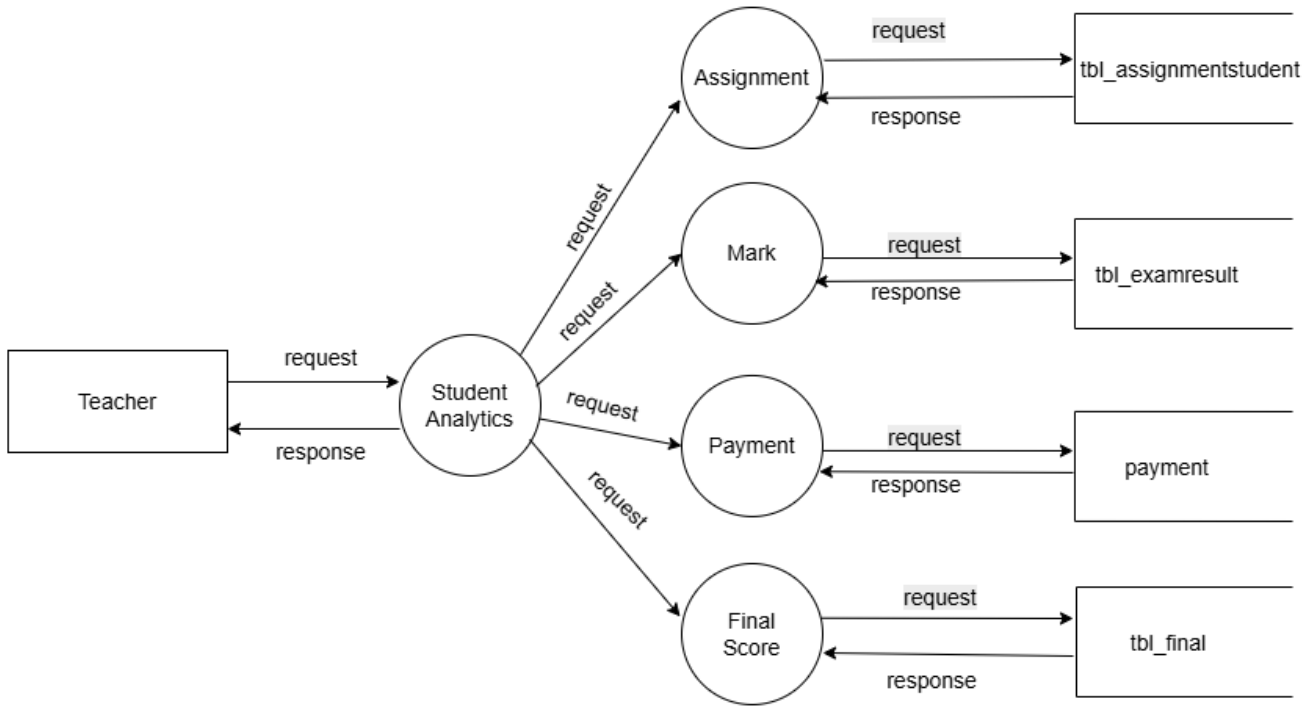
LEVEL 2.0 DFD



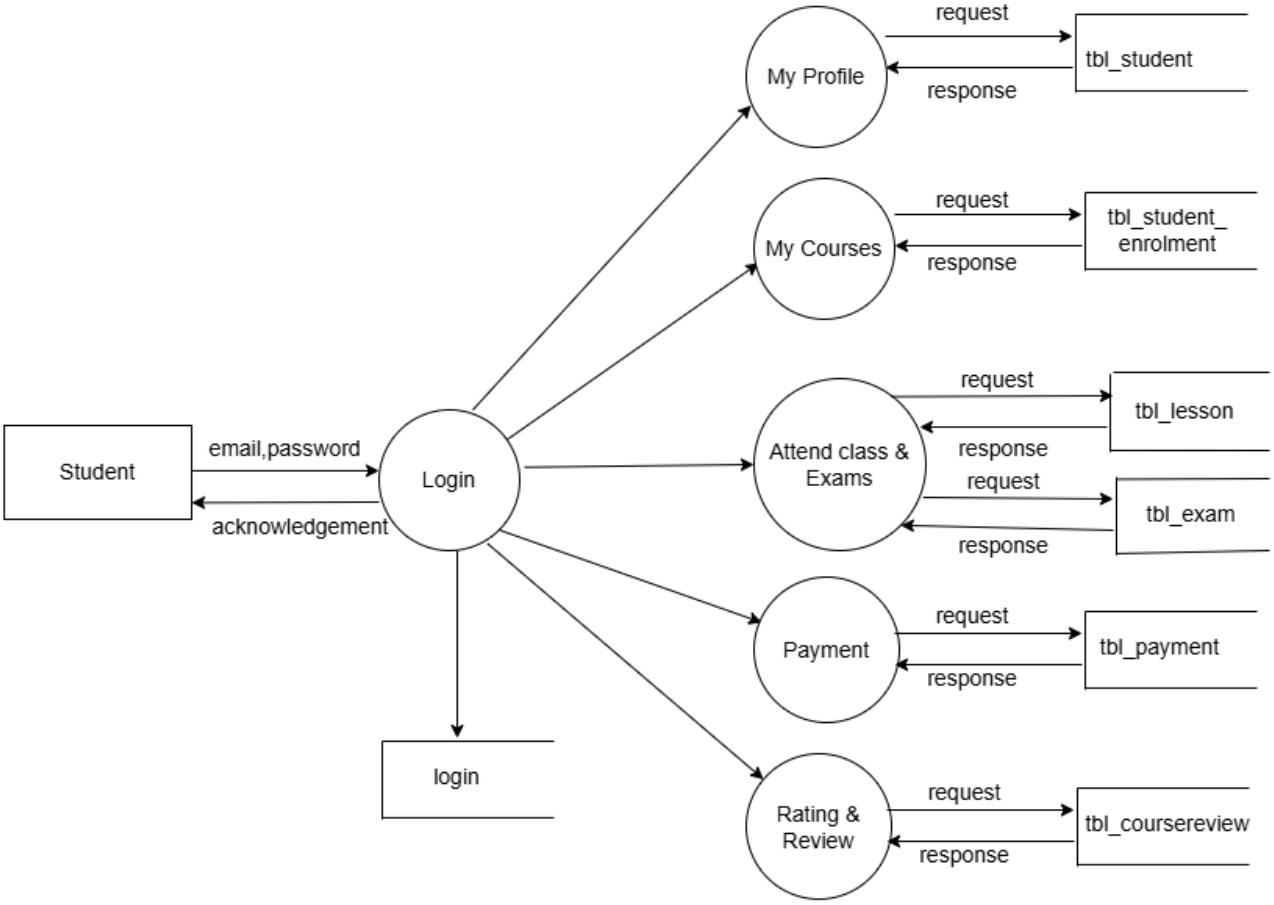
Level 2.1 DFD



Level 2.2 DFD



LEVEL 3.0 DFD


Student
email,password
acknowledgement
Login
login
My Profile
request
response
tbl_student
My Courses
request
response
tbl_student_
enrolment
Attend class &
Exams
request
response
tbl_lesson
request
response
tbl_exam
Payment
request
response
tbl_payment
Rating &
Review
request
response
tbl_coursereview

LEVEL 3.1 DFD

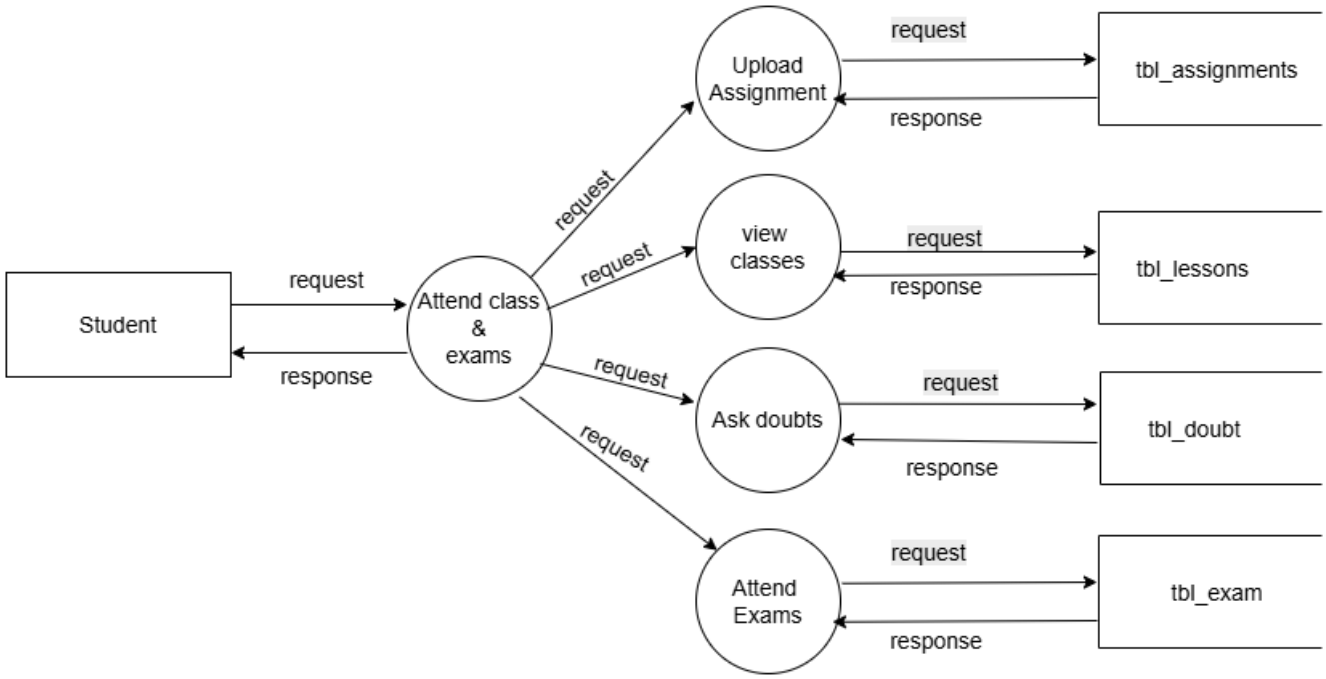

Student
request
response
Attend class & exams
request
request
request
request
Upload Assignment
view classes
Ask doubts
Attend Exams
request
response
request
response
request
response
request
response
tbl_assignments
tbl_lessons
tbl_doubt
tbl_exam

TABLE DESIGN

Sl. No	Table Name	Description
1	login	To store login credentials and user roles
2	student	To store student profile details
3	teacher	To store teacher profile details
4	stream	To store education streams
5	course	To store course details
6	batch	To store batch details
7	teacherbatch	To assign teachers to batches
8	student_enrolment	To store student enrolment details
9	payment	To store payment details
10	exam	To store exam details
11	questions	To store exam questions
12	options	To store question options
13	examquestions	To link questions with exams
14	examresult	To store exam results
15	assignment	To store assignments
16	assignmentstudent	To store assignment submissions
17	lesson	To store course lessons
18	doubt	To store student doubts
19	course_review	To store course reviews

The general theme behind a database is to handle information as an integrated whole. A database is a collection of interrelated data stored with minimum redundancy to serve quick access and effective storage. The database is a collection of stored data organized in such a way that all the data requirements are satisfied by the database. The aim of database design is to improve the existing system situation. A number of database files were designed to hold the data requirements for running their systems. Here we have 19 major tables, described below

Table No: 1

Table Name: login
Description: Stores login credentials and user roles.
Primary Key: id
Foreign Key: None

Sl.No	Field Name	Data Type	Description
1	id	BigInt	Unique login ID
2	email	Varchar(254)	Stores user email
3	password	Varchar(255)	Stores encrypted password
4	usertype	Varchar(20)	Defines user role (Admin/Teacher/Stud)

Table No: 2

Table Name: student
Description: Stores student profile information.
Primary Key: id
Foreign Key: login_id → login(id)

Sl.No	Field Name	Data Type	Description
1	id	BigInt	Unique student ID
2	firstname	Varchar(100)	First name
3	lastname	Varchar(100)	Last name
4	phone	Varchar(15)	Contact number
5	address	LongText	Student address
6	profilepic	Varchar(100)	Profile image
7	login_id	BigInt	Reference to login table

Table No: **3**

Table Name: teacher

Description: Stores teacher profile details.

Primary Key: id

Foreign Key: login_id → login(id)

Sl.No	Field Name	Data Type	Description
1	id	BigInt	Unique teacher ID
2	firstname	Varchar(100)	First name
3	lastname	Varchar(100)	Last name
4	phone	Varchar(15)	Contact number
5	qualification	Varchar(100)	Teacher qualification
6	profilepic	Varchar(100)	Profile image
7	login_id	BigInt	Reference to login table

Table No: **4**

Table Name: stream

Description: Stores different education streams.

Primary Key: id

Foreign Key: None

Sl.No	Field Name	Data Type	Description
1	id	BigInt	Unique stream ID
2	stream_name	Varchar(100)	Name of stream

Table No: 5

Table Name: course

Description: Stores course information**.**

Primary Key: id

Foreign Key: stream_id → stream(id)

Sl.No	Field Name	Data Type	Description
1	id	BigInt	Unique course ID
2	course_name	Varchar(150)	Course name
3	description	LongText	Course details
4	price	Decimal	Course fee
5	stream_id	BigInt	Reference to stream table

Table No: 6

Table Name: batch
Description: Stores batch details for courses.
Primary Key: id
Foreign Key: course_id → course(id)

Sl.No	Field Name	Data Type	Description
1	id	BigInt	Unique batch ID
2	batch_name	Varchar(100)	Batch name
3	start_date	Date	Batch starting date
4	end_date	Date	Batch ending date
5	course_id	BigInt	Reference to course

Table No: 7

Table Name: teacherbatch
Description: Assigns teachers to batches.
Primary Key: id
Foreign Key: teacher_id → teacher(id), batch_id → batch(id)

Sl.No	Field Name	Data Type	Description
1	id	BigInt	Unique ID
2	teacher_id	BigInt	Reference to teacher
3	batch_id	BigInt	Reference to batch

Table No: 8

Table Name: student_enrolment
Description: Stores course enrolment details of students.
Primary Key: id
Foreign Key: student_id → student(id), batch_id → batch(id)

Sl.No	Field Name	Data Type	Description
1	id	BigInt	Enrolment ID
2	student_id	BigInt	Reference to student
3	batch_id	BigInt	Reference to batch
4	enrol_date	Date	Enrolment date

Table No: 9

Table Name: payment
Description: Stores payment details for course enrolment.
Primary Key: id
Foreign Key: enrolment_id → student_enrolment(id)

Sl.No	Field Name	Data Type	Description
1	id	BigInt	Payment ID
2	enrolment_id	BigInt	Reference to enrolment
3	amount	Decimal	Payment amount
4	payment_date	Date	Payment date
5	status	Varchar(50)	Payment status

Table No: 10

Table Name: exam
Description: Stores exam details.
Primary Key: id
Foreign Key: course_id → course(id)

Sl.No	Field Name	Data Type	Description
1	id	BigInt	Exam ID
2	exam_name	Varchar(150)	Exam name
3	total_marks	Int	Maximum marks
4	exam_date	Date	Exam date
5	course_id	BigInt	Reference to course

Table No: 11

Table Name: questions
Description: Stores exam questions.
Primary Key: id
Foreign Key: exam_id → exam(id)

Sl.No	Field Name	Data Type	Description
1	id	BigInt	Question ID
2	exam_id	BigInt	Reference to exam
3	question_text	LongText	Question content

Table No: 12

Table Name: options
Description: Stores options for each question.
Primary Key: id
Foreign Key: question_id → questions(id)

Sl.No	Field Name	Data Type	Description
1	id	BigInt	Option ID
2	question_id	BigInt	Reference to question
3	option_text	Varchar(255)	Option text
4	is_correct	Boolean	Indicates correct option

Table No: 13

Table Name: examquestions
Description: Links questions with exams.
Primary Key: id
Foreign Key: exam_id → exam(id), question_id → questions(id)

Sl.No	Field Name	Data Type	Description
1	id	BigInt	Unique ID
2	exam_id	BigInt	Reference to exam
3	question_id	BigInt	Reference to question

Table No: 14

Table Name: examresult
Description: Stores exam results of students.
Primary Key: id
Foreign Key: exam_id → exam(id), student_id → student(id)

Sl.No	Field Name	Data Type	Description
1	id	BigInt	Result ID
2	exam_id	BigInt	Reference to exam
3	student_id	BigInt	Reference to student
4	marks_obtained	Int	Marks scored
5	result_status	Varchar(50)	Pass/Fail

Table No: 15

Table Name: assignment
Description: Stores assignments for courses.
Primary Key: id
Foreign Key: course_id → course(id)

Sl.No	Field Name	Data Type	Description
1	id	BigInt	Assignment ID
2	title	Varchar(150)	Assignment title
3	description	LongText	Assignment details
4	due_date	Date	Submission deadline
5	course_id	BigInt	Reference to course

Table No: 16

Table Name: assignmentstudent
Description: Stores assignment submissions by students.
Primary Key: id
Foreign Key: assignment_id → assignment(id), student_id → student(id)

Sl.No	Field Name	Data Type	Description
1	id	BigInt	Submission ID
2	assignment_id	BigInt	Reference to assignment
3	student_id	BigInt	Reference to student
4	submission_file	Varchar(255)	Uploaded file
5	marks	Int	Marks obtained

Table No: 17

Table Name: lesson
Description: Stores lessons under courses.
Primary Key: id
Foreign Key: course_id → course(id)

Sl.No	FieldName	Data Type	Description
1	id	BigInt	Lesson ID
2	lesson_title	Varchar(150)	Lesson title
3	video_link	Varchar(255)	Video URL
4	description	LongText	Lesson details
5	course_id	BigInt	Reference to course

Table No: 18

Table Name: doubt
Description: Stores doubts raised by students.
Primary Key: id
Foreign Key: student_id → student(id), course_id → course(id)

Sl.No	Field Name	Data Type	Description
1	id	BigInt	Doubt ID
2	student_id	BigInt	Reference to student
3	course_id	BigInt	Reference to course
4	doubt_text	LongText	Student question
5	reply	LongText	Teacher response

Table No: 19

Table Name: course_review
Description: Stores reviews and ratings for courses.
Primary Key: id
Foreign Key: student_id → student(id), course_id → course(id)

Sl.No	Field Name	Data Type	Description
1	id	BigInt	Review ID
2	student_id	BigInt	Reference to student
3	course_id	BigInt	Reference to course
4	rating	Int	Rating (out of 5)
5	review_text	LongText	Student feedback

SYSTEM DESCRIPTION

1. User Registration and Verification:

• Students can register in the system by providing basic details such as name, email, contact number, and address through an online registration form.
• The system allows immediate access to students after successful registration and login, ensuring a smooth and user-friendly onboarding process.
• Teacher accounts are managed under administrative control to maintain the quality and authenticity of educational content.
• The admin is responsible for adding and managing teacher accounts, ensuring that only authorized individuals are given teaching privileges within the platform.

2. Login and Authentication:

• Registered users can log in using valid credentials such as email and password.
• The system ensures secure authentication for Students, Teachers, and Admin.
• Admin has the authority to manage login access and deactivate accounts if required.
• Role-based access control ensures that each user accesses only their respective features.

3. User Profiles:

• Each user has an individual profile containing their personal and academic details.
• Students and teachers can update their profile information such as contact details and profile picture.
• Teachers can include additional details like qualifications and experience.
• Users can change their passwords securely from their profile section.

4. Course and Batch Management:

• Courses are organized under different streams and managed by the admin.
• Each course can have multiple batches such as Batch A, Batch B, etc.
• Admin assigns teachers to specific batches based on course requirements.
• Students can enroll in available batches based on their course selection.

5. Course Enrollment and Access:

• Students can enroll in courses and join specific batches.
• Once enrolled, students can access study materials such as videos, notes, and PDFs uploaded by teachers.
• The system tracks enrolled courses and provides structured learning access.
• Students can view their enrolled courses and learning progress through their dashboard.

6. Study Materials and Lesson Management:

• Teachers can upload study materials including videos, documents, and notes for each course.
• Lessons are organized in a structured manner for easy understanding.
• Students can access materials anytime after enrollment.
• This ensures a flexible and continuous learning experience.

7. Assignment Management:

• Teachers can assign tasks and assignments to students within a course.
• Students are required to submit assignments within the given deadline.
• Teachers can evaluate submitted assignments and provide marks.
• Assignment completion is mandatory for exam eligibility.

8. Payment Module:

• Students must complete course payment before accessing full features.
• The system securely handles payment transactions and stores payment records.
• Payment status is tracked and displayed in the student dashboard.
• Only students who complete payment are eligible to attend exams.

9. Examination System:

• Admin manages the examination process for each course.
• Exams are conducted after completion of assignments and payment.
• Questions and options are managed within the system.
• Students can attend exams and receive marks based on their performance.

10. Result and Performance Evaluation:

• The system stores exam results and assignment marks of students.
• An automatic process combines assignment marks and exam marks to calculate overall performance.
• Students can view their results and performance through dashboards.
• Teachers can monitor student performance batch-wise.

11. Certification System:

• After successful completion of the course, students receive certificates.
• Certificates are generated with a unique identification number.

• Only students who complete assignments, payment, and exams are eligible.
• Certificates can be downloaded from the student dashboard.

12. Dashboard and Analytics:

• Each user has a dedicated dashboard for easy access to features.
• Students can view courses, progress, marks, and payments.
• Teachers can track student performance and manage content.
• Admin dashboard includes system statistics, user management, and graphical reports.

13. Doubt Management and Interaction: Students can raise doubts related to lessons and courses.
• Teachers can respond to student queries within the system.
• This improves interaction and enhances learning clarity.
• The system ensures continuous communication between students and teachers.

14. User Management (Admin):

• Admin manages students and teachers, including adding, updating, and removing users.
• Admin controls course creation, batch allocation, and teacher assignment.
• Payment tracking and system monitoring are handled by admin.
• Ensures smooth functioning and maintenance of the platform.

SYSTEM TESTING AND IMPLEMENTATION

SYSTEM TESTING

Software testing is an important phase in system development that ensures the quality, reliability, and performance of the application. It involves executing the system with test data to identify errors and verify whether the system meets the specified requirements. Testing helps in detecting issues at different stages of development and ensures that the final product functions correctly.

In NovaLearn, testing was carried out at various levels to ensure that all modules such as user management, course handling, assignments, and payments work efficiently. A series of tests were conducted before the system was made ready for user acceptance, ensuring that the system performs as expected.

TYPES OF TESTING

- Unit Testing
- Integration Testing
- Validation Testing
- Output Testing

UNIT TESTING

Unit testing focuses on verifying individual modules of the system to ensure that each unit performs correctly. Each module is tested separately to identify errors in logic, data handling, and functionality. This testing is usually performed during the development phase itself.

In NovaLearn, unit testing was conducted on modules such as login, registration, course management, and assignment submission. Each form and functionality was tested to ensure that data is correctly stored in the database and that the expected output is generated without errors.

INTEGRATION TESTING

Integration testing is performed after unit testing, where different modules are combined and tested as a whole. The objective is to identify errors that occur due to interaction between modules, such as data flow issues or incorrect linking.

In NovaLearn, modules like student enrolment, payment, assignments, and exams were integrated and tested together. Issues related to database relationships and data flow between modules were identified and corrected, ensuring smooth interaction across the system.

VALIDATION TESTING

Validation testing ensures that the system meets user requirements and performs as expected in real-world scenarios. It verifies whether the developed system satisfies the functional requirements defined during the design phase.

In NovaLearn, validation testing was carried out by checking whether all features such as course access, assignment submission, exam eligibility, and result generation function correctly. The system was tested and confirmed to meet user expectations and requirements.

OUTPUT TESTING

Output testing is performed to verify whether the system produces the correct output in the desired format. It ensures that reports, results, and displayed data are accurate and user-friendly.

In NovaLearn, output testing was done for dashboards, exam results, payment details, and certificates. The outputs were checked for correctness, clarity, and format, and were found to be satisfactory according to user requirements.

SYSTEM IMPLEMENTATION

System implementation is the process of converting the designed system into a working application. It involves planning, testing, and deploying the system so that users can start using it effectively. Proper implementation ensures that the system operates smoothly and meets user expectations.

In NovaLearn, implementation was carried out after successful testing of all modules. The system was deployed in a controlled environment where users could access features such as course enrollment, learning materials, assignments, and exams. Proper planning ensured smooth transition and minimized errors during deployment.

TYPES OF IMPLEMENTATION

- Implementation of a new computer system to replace manual processes.
- Implementation of a system to replace an existing system.
- Implementation of a modified system with improved features.

CONCLUSION

In conclusion, the project NovaLearn represents a comprehensive and efficient digital learning platform designed to connect students, teachers, and administrators within a unified and interactive environment. By integrating key modules such as course management, content delivery, assignments, examinations, payments, and certification, NovaLearn provides a structured and user-friendly system for online education.

The administrative functionalities enable the admin to effectively manage students, teachers, courses, and batches while ensuring smooth system operation. The platform supports organized learning through structured courses, lesson uploads, and assignment management. The integrated examination and evaluation system ensures accurate performance assessment, while the payment module maintains secure and transparent transactions.

One of the key strengths of NovaLearn is its automated evaluation and certification process, where assignment marks and exam results are combined to generate final performance outcomes. Students can track their progress through dashboards and download certificates with unique identification numbers upon successful completion. Overall, NovaLearn enhances accessibility, efficiency, and transparency in the learning process, creating a reliable and scalable solution for modern digital education.

BIBLIOGRAPHY

• **System Analysis and Design**
Elias M. Awad
Second Edition, 1999
Galgotia Publications, New Delhi

• **Database System Concepts**
Abraham Silberschatz, Henry F. Korth, S. Sudarshan
Fourth Edition
McGraw-Hill Publications

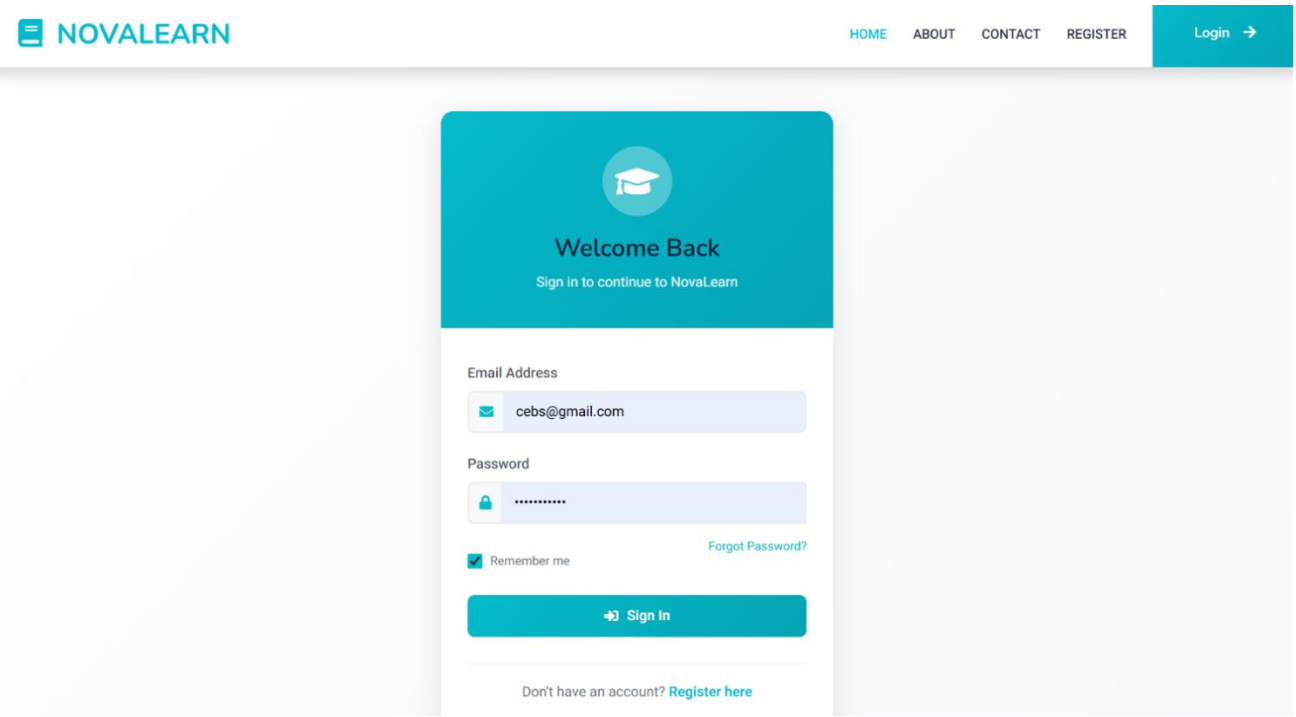
• **Web Programming using Python and Django**
Benny Jacob
Kalyani Publications

• **HTML, CSS and JavaScript Web Publishing**
Laura Lemay
Sams Publishing

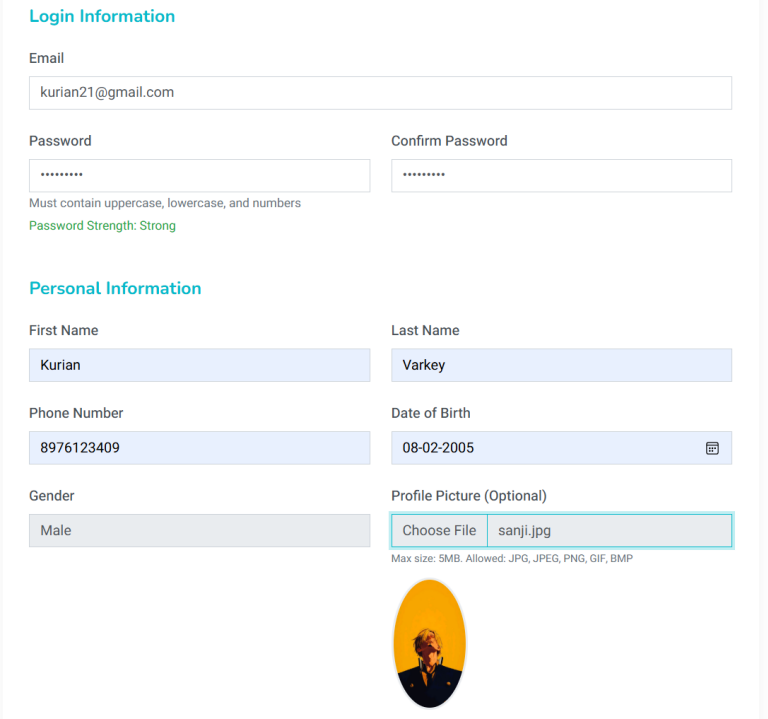
APPENDEX

Input Forms

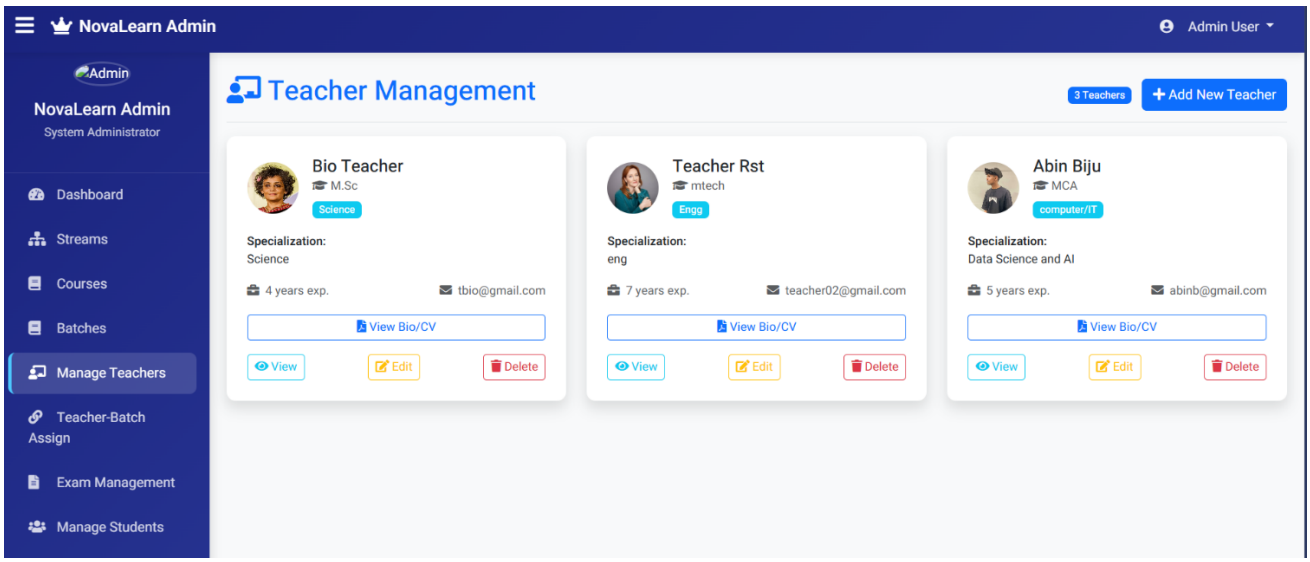
Form-1-Login



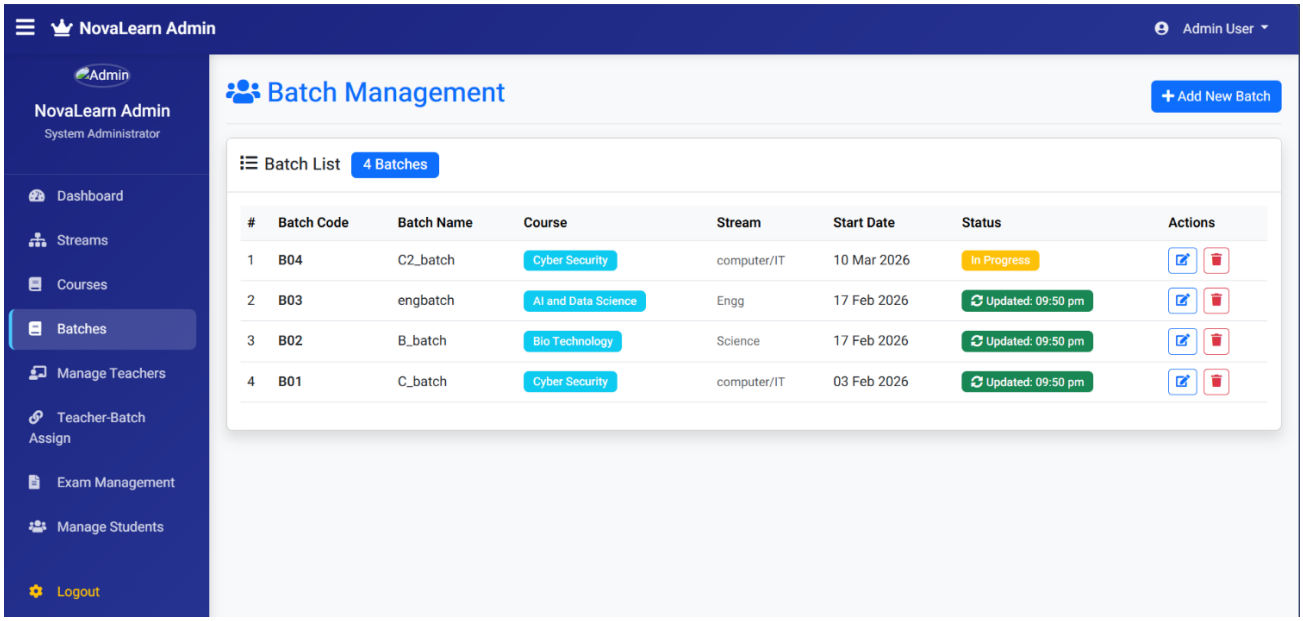
Form-2-Student registration



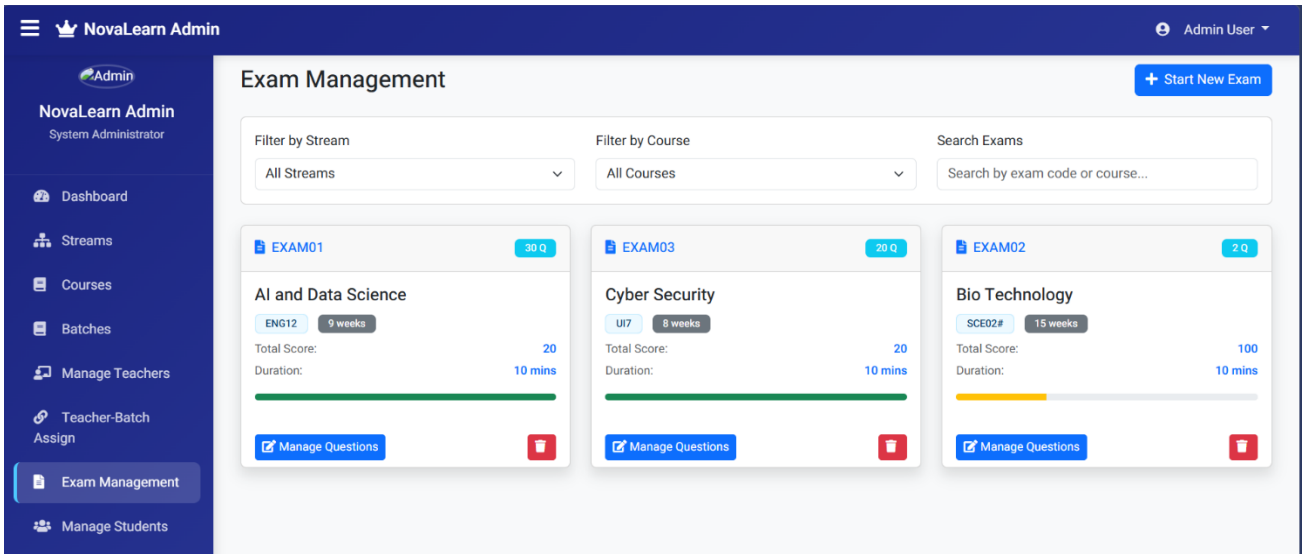
Form-3-Admin Teacher management



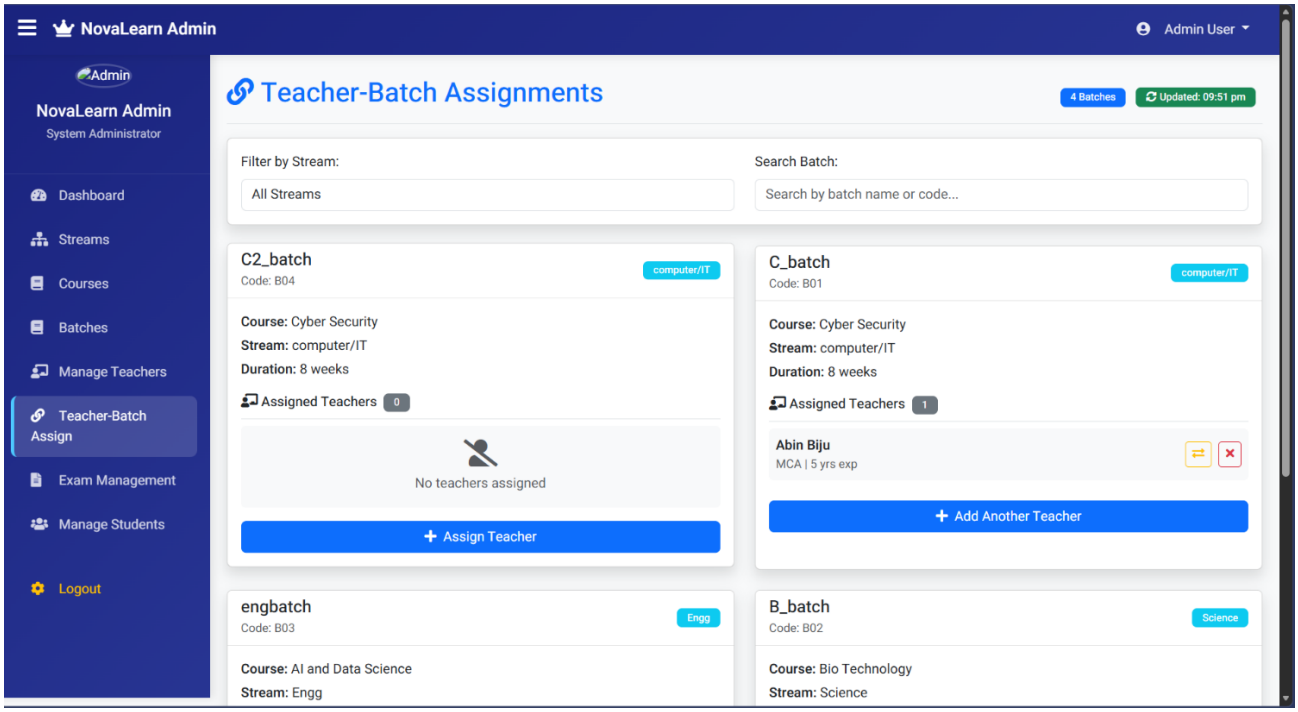
Form-4-Admin Batch Management



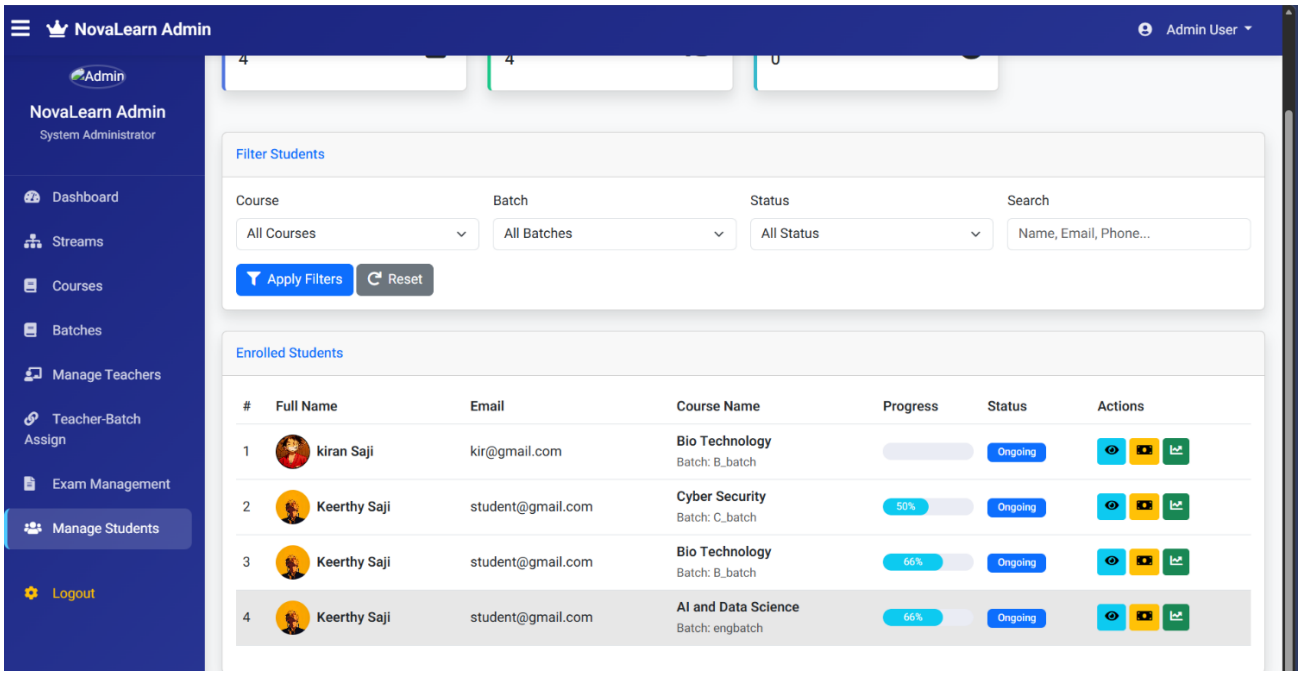
Form-5-Admin exam management



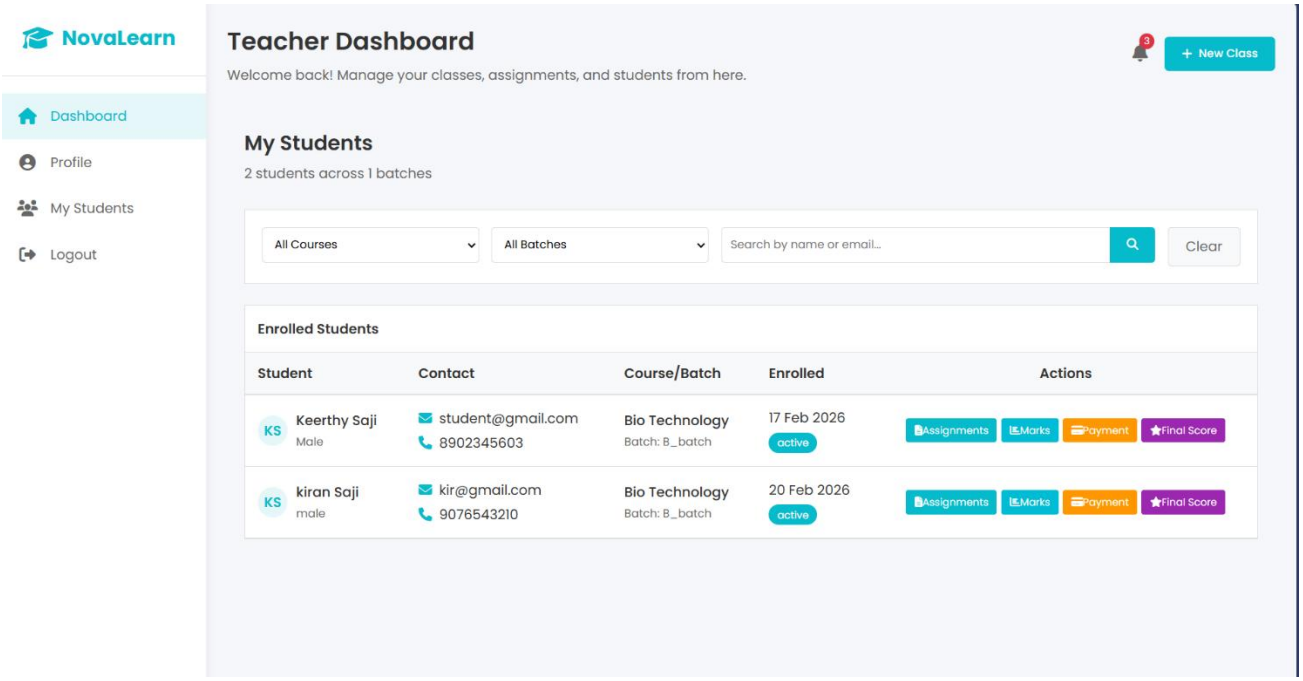
Form-6-Admin Teacher-batch management



Form-7-Admin Student Management

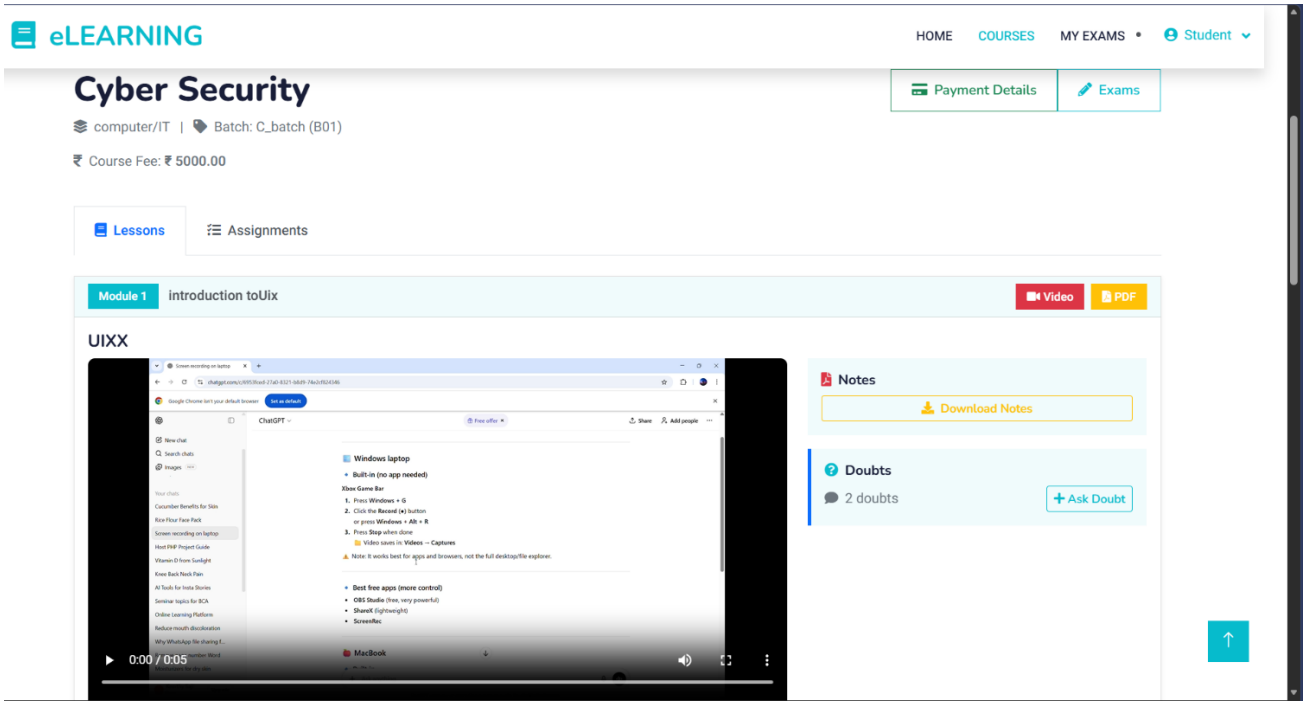


Form-8-Teacher student evaluation

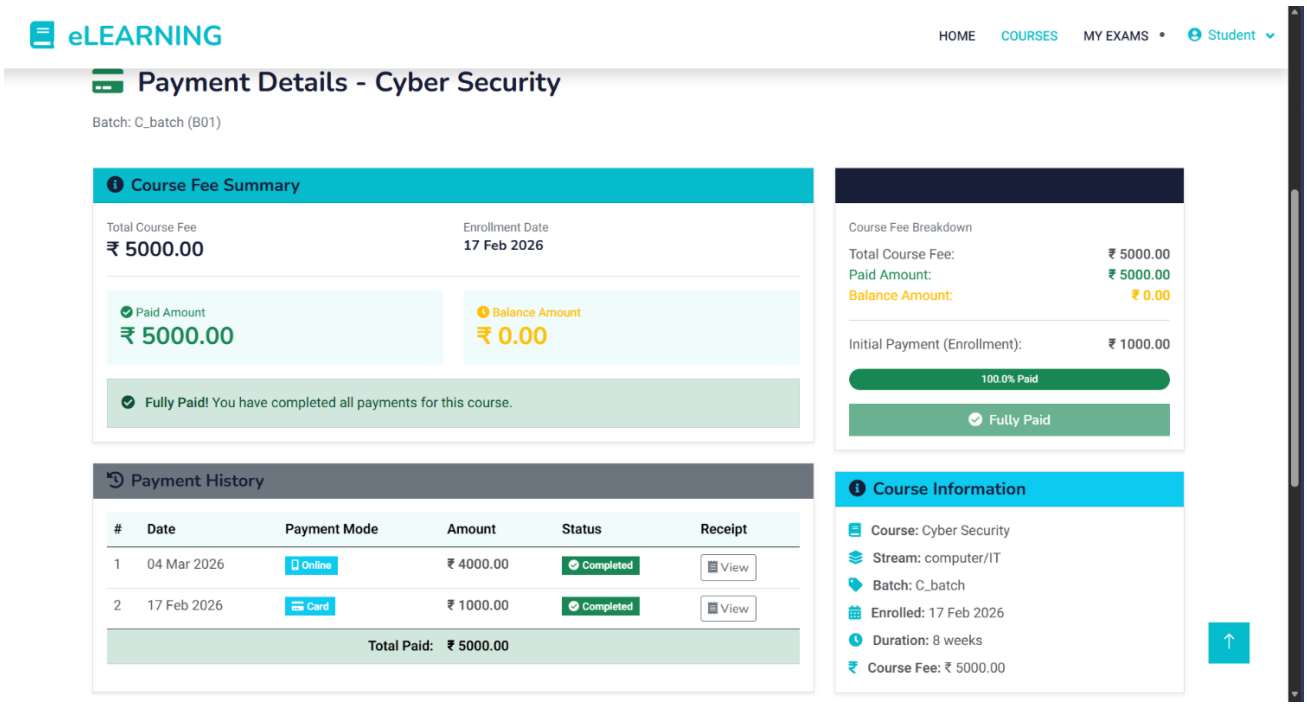
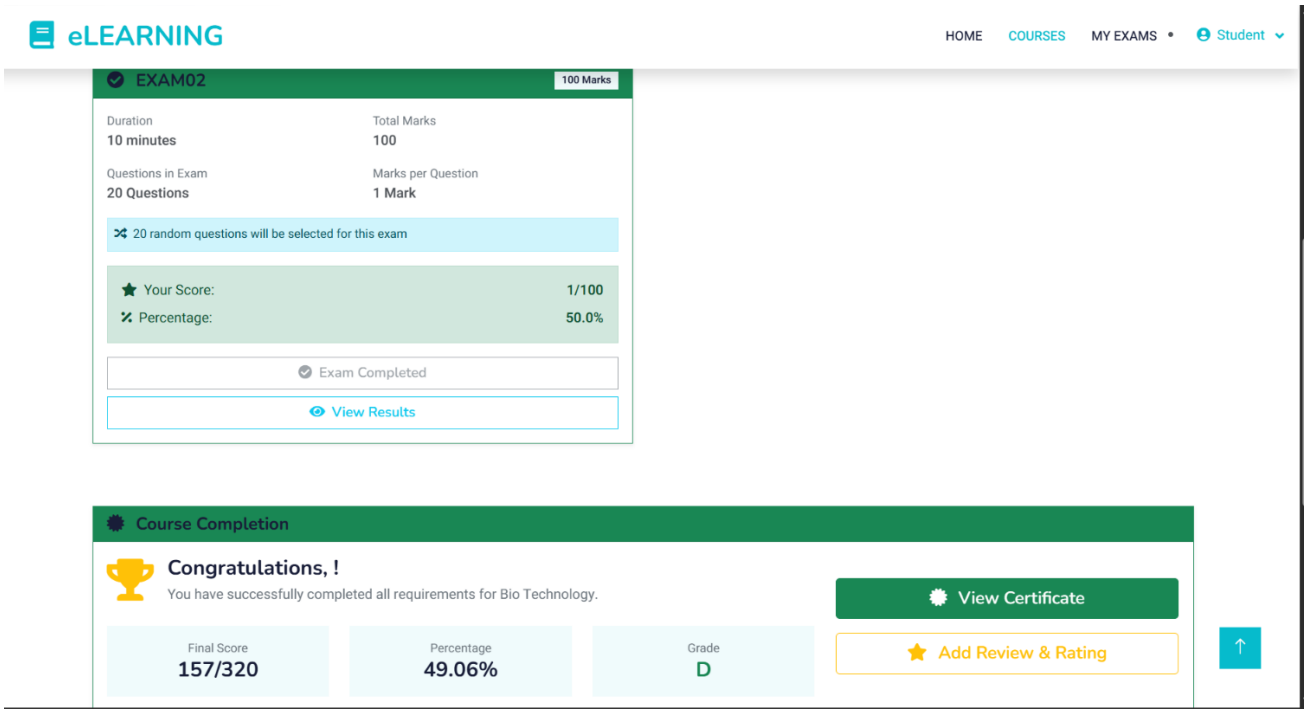


Output Forms

Form-9-Student Lesson and Assignment management



Form-10-Student Payment Details



Form-11-Student Exam and Certificate access